REFERENCE DOCUMENT

Introduction

A concise project overview aligned with the judging criteria

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SECTION

Introduction

TraversalAI is a multi-agent AI and edge-runtime project designed to turn a simple user request into an organized, trackable outcome. Instead of presenting AI as a black-box chatbot, the project exposes a full operating flow: user input, gateway routing, agent decision-making, tool execution, and saved session continuity.

SECTION

Problem and Theme Fit

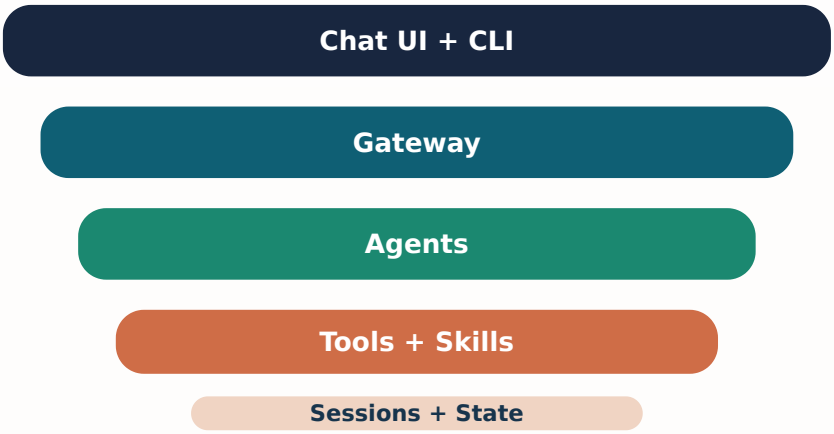
The judging rules emphasize real-world value inside the tourism and hospitality theme. TraversalAI supports that framing by acting as an intelligent service layer for visitor guidance, service coordination, multilingual assistance, and task handling across multiple channels.

- It can guide visitors before, during, and after their trip.
- It can coordinate service actions through specialized agents.
- It can present a clear AI + IoT story when hosted on Raspberry Pi 5 with microphone and speaker.

SECTION

How the Prototype Works

The prototype routes every interaction through the TraversalAI gateway, allowing judges to see a structured system rather than a disconnected demo. Agents, models, tools, and skills are visible parts of the execution path.



SECTION

What Makes It Judge-Ready

- **Design:** clear user flow and system boundaries.
- **Innovation:** multi-agent orchestration instead of one undifferentiated assistant.
- **Electronics:** Raspberry Pi 5, microphone, and speaker create a tangible AI + IoT setup.
- **Programming:** gateway-backed execution, session handling, and tool integration are implemented in code.
- **Presentation:** a live path from request to result can be shown quickly.
- **Impact:** the system can be framed around real tourism and hospitality pain points.

Problem → **Request** → **Decision** → **Execution** → **Outcome**

SECTION

Presentation Guidance

```
1. Open with the tourism or hospitality problem.
2. Show the hosted prototype environment.
3. Run one short scenario from request to outcome.
4. Point to the active agent, tool, or skill during execution.
5. End with the expected user impact and why the solution is scalable.
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This English introduction is a styled, reviewer-friendly version derived from the content direction in P_10.pdf and aligned to the official judging guidebook.